NORTHERN CARIBBEAN UNIVERSITY
A Seventh-day Adventist Institution

Name: Ray-Shaun Bourne

Student ID#: 20211553

Lecturer: Miss. Tora Simpson

Course: Engineering and Orientation Ethics (ENGR110)

Due Date: April 14, 2022

Assignment: Final Report

Table of Contents

Acknowledgement

I would like to thank my teacher/instructor who gave me this opportunity to work on this project and giving me the basis on how the project should be done. I got to learn a lot from this project about how the hovercraft works, how wiring connection works, the history of the hovercraft and how to improvise when something goes wrong with a project that is being worked on. I would like to extend my heartfelt thanks to my family because without their help this project would not have been successful. I would like to thank my dear friends who have would have also helped me as well and finally, I would like to thank god for blessing me throughout this journey with being able to get the things needed for this project to be done.

Abstract/Executive Summary

For many years there were many efforts to reduce the element of drag between moving vehicles. A hovercraft is new means of transportation. The concept of the hovercraft was born when engineers developed a design used for experiments to reduce the drag force of a moving vehicle. The idea was to use a cushion of air between the boat and the water to reduce the drag force. The aim of this project is to design and build (design using software such as AutoCAD and Fusion360) a small working model of a hovercraft with a system that allows the hovercraft to propel horizontally covering a distance of four yards. Usually, a hovercraft is designed to be used with a vertical propulsion system to produce lift and thrust forces. This model can perform the basic function of a hovercraft to travel on a surface but the model used was limited to only travelling on land surfaces. The model produces a reasonable amount of air cushion to lift its body and carry a small load, and at the same time the model produces some form of thrust to move. A functioning hovercraft model needs to be design in consideration of the horizontal propulsion system to reduce drag force, minimize the lift force and improve thrust force. The outcome expected for this project is to design and build a small working model of a hovercraft with a system that allows the hovercraft to propel horizontally covering a distance of four yards.

Introduction

A hovercraft is a vehicle that files like a plane but can float like a boat, can drive like a car but can traverse through things that a car couldn't normally traverse through. A hovercraft is also sometimes called an air cushion vehicle because it can hover over, or move across land and water surfaces while being held off from the surface by a cushion of air. Hovercrafts can move across all types of surfaces including grass, mud, quicksand, sand, water and ice. Hovercrafts prefer a gentle terrain although they are capable of climbing slopes of up to 20%, depending on what the surface is. Hovercrafts are used for many applications where people and equipment need to travel at high speeds over various surfaces and be able to load and unload on land.

Methodology/Procedure

To develop a small-scale hovercraft, there were 5 main criteria requirements made for the designs:

1. Cost Effectiveness

2. Weight

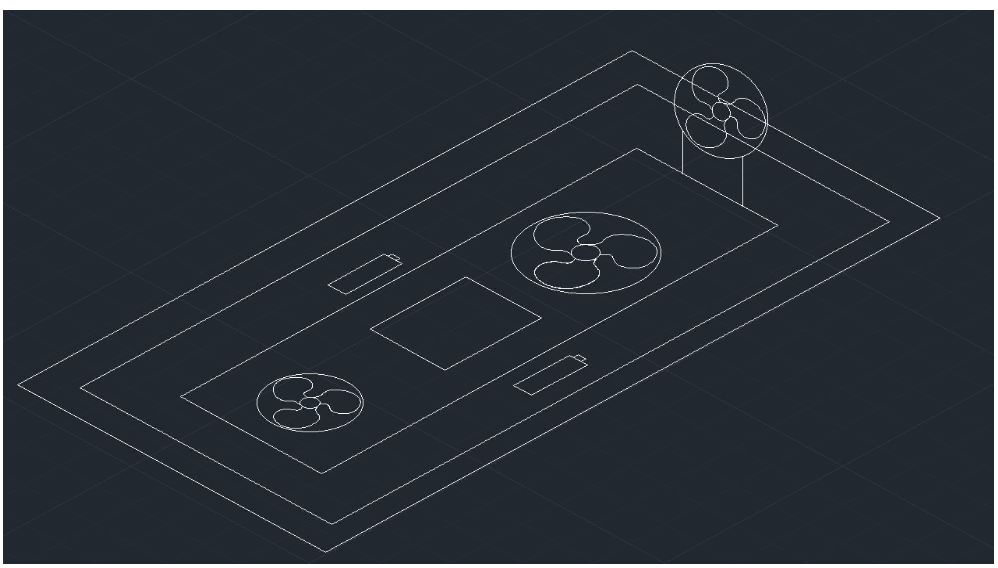
3.Travels 4 – 6ft

4. Simplicity

5. Power Consumed

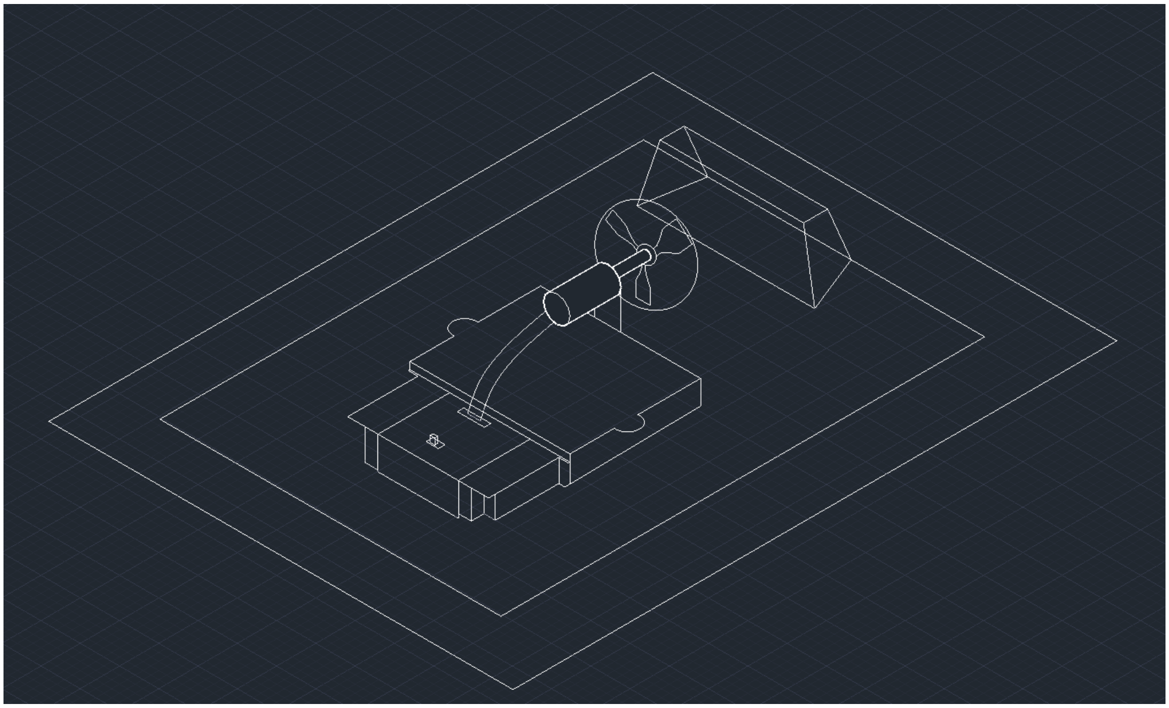
Based off the criteria requirements, 2 hovercraft designs with the horizontal propulsion system were made through the use of AutoCAD. They are as shown below:

Design 1



Design 1 was drawn using AutoCAD. The design uses three fan blades, a control panel (which contains the motor and the switch in the middle) and powered by two batteries. Also containing the skirt (which is the outside region of the design.)

Design 2



Design 2 was drawn using AutoCAD. The design uses one main fan blade which propels the hovercraft horizontally/forward. In the middle of the design is the control panel which is powered by four batteries, contains a switch, motor stand to hold the motor and the motor itself.

Criteria Matrix.

The criteria matrix was based on the concept of existing real hovercrafts and RC model hovercrafts.

Evaluation Matrix:	Possible Effectiveness Score	Score for Design 1	Score for Design 2
1. Cost Effectiveness	40	25	30
2. Weight	20	10	10
3. Travels 4 – 6 ft	20	10	15
4. Simplicity	10	10	5
5. Power Consumed	10	8	10
Total effectiveness score	100	63	70

Based on the Criteria Matrix, design 2 was selected. For cost effectiveness design 1 would be more expensive than design 2 because of the extra cost of getting three fan blades and possibly 3 dc motors as compared to design 2 which only utilizes one fan blade and one dc motor. The only disadvantage for cost effectiveness with design 2, it uses four batteries compared to design 1 it uses two batteries. In terms of weight Design 1 and Design 2 match up. In terms of travelling 4 – 6 ft. both designs would travel in that distance however, design 2 based on the battery power would more than likely travel farther than 4-6 ft. Based on simplicity design 1 has a simpler build compared to design 2. As for the Power Consumed design 1 would consume less power compared to design 2.

Materials and Components:

1. Base panel (Black Cardboard)

2.Plastic Skirt

3. Duct Template (Plastic)

4. Battery Compartment with switch and wires

5. Battery Compartment Cover

6. Motor Stand

7. Motor Cover

8.Attachment Bar

9. Propeller (Fan Blade)

10. Motor with wires

11. Screws.

12. Terminal Cap.

13. Double-sided Adhesive Tape.

Photos of few of the materials and Components.

Selection of Material for Skirt.

The skirt are air bags inflated by air and fitted around the hovercraft holding air under the hovercraft and thus upon a cushion of air. It enables the hovercraft to obtain a greater hover height. As for the model built, the material used for the skirt was chosen to be plastic. This is because plastic is the most basic and easiest material to use to build the skirt as it can be easily cut (for the places where the air needs to be sucked in so the skirt can inflate) and easily taped to the base panel.

Selection of Material for Fan.

To produce the lifting and thrusting force, the propeller (fan blade) plays the main role. One main propeller was chosen for the hovercraft model. The reason for choosing one main propeller is that it produces a high speed of air circulation which inflates the skirt and makes the hovercraft move (however it was tested and depending on how much charge the battery has, the high speed of air circulation which inflates the skirt may vary).

Process of Building the Hovercraft.

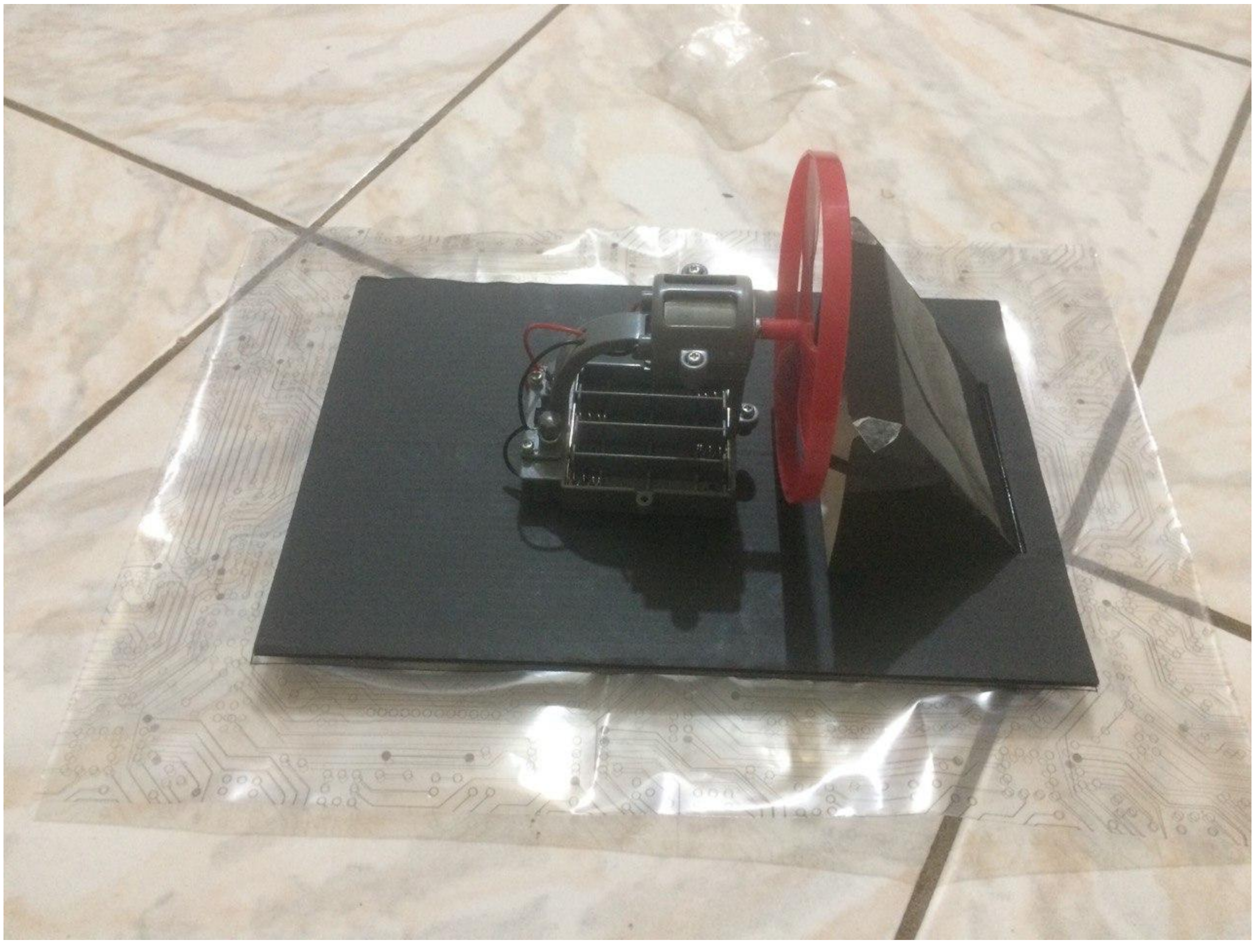
Double-sided adhesive tape was applied along the edges of the base panel as this would be placed on the skirt. It should also be mentioned some holes were drilled in the base panel (as seen in the pictures above of the materials/components) as the battery compartment with the switch and wires would be placed on the base panel to hold it in place (Along with the screws). The double-sided adhesive tape was also applied to the duct template and then placed on the base panel. The skirt was prepared as the skirt is a major part of the hovercraft model. To make the skirt, plastic was used and a pair of scissors was used to cut the shape out of the skirt. The skirt consisted of two layers of plastic. The bottom part of the skirt was cut in order for the air to come in which would inflate the skirt. The base panel with the edges that were taped were placed on the skirt carefully.

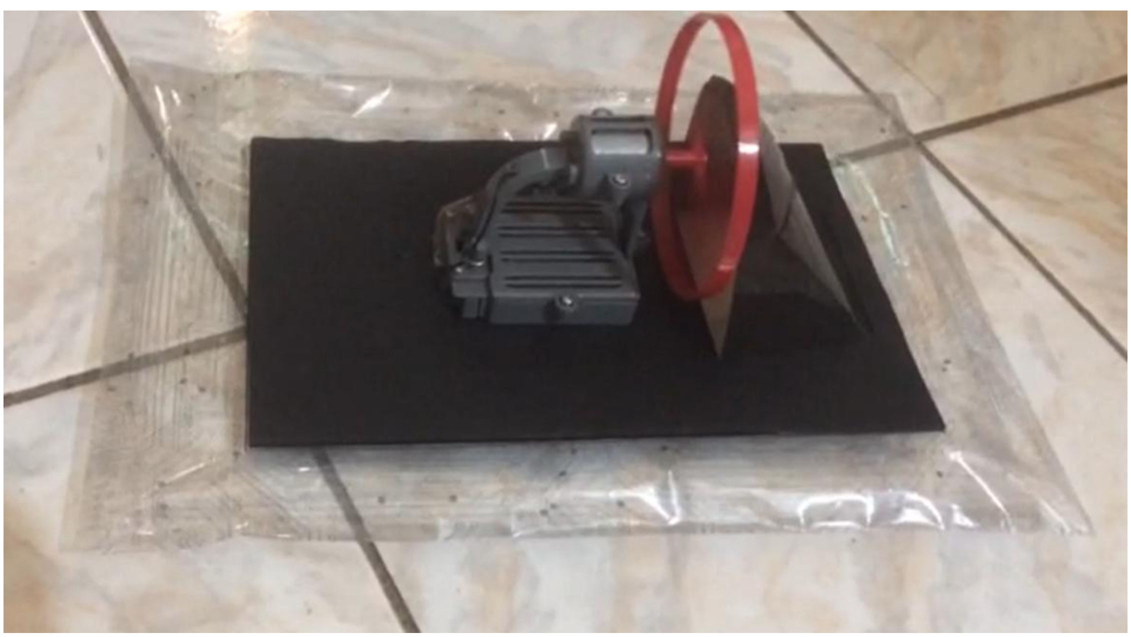
Before placing the battery compartment on the base panel, the motor stand was placed on the battery compartment and secured with three of the screws. The motor was then placed inside the motor stand with the wires pointing towards the curved leg of the stand. The wires were placed through the holes of the motor stand and the motor cover was then placed on the motor and secured with two screws. The wires from the motor itself were then twisted together with the wires from the battery compartment with their respective colour. The wires were then placed in the terminal holes of the battery compartment and the secured with the two terminal caps. Batteries were then placed in the battery compartment and the battery compartment cover was placed over the batteries to hold the batteries in place and secured with screws (shown in picture below). With everything connected, the battery compartment was then placed on the base panel were the holes were drilled in. To hold the battery compartment in place (with everything else connected), the two attachment bars were placed on the bottom of the base panel (were the same holes were drilled) and the secured with screws (Shown in the pictures below). The propeller pushed firmly onto the motor spindle.

Batteries placed in the battery compartment then the battery cover placed to hold the batteries in place secured with screws.

Attachment Bars placed to hold the battery compartment in place.

Complete design of hovercraft.





Results

After testing was done, it was concluded that the hovercraft does work as intended but it was moving at a slow rate at first. Troubleshooting was done and it was deemed because of the charge on the batteries being very low the hovercraft was moving slow. As a results new batteries were placed into the model and the hovercraft began to move at the rate it would originally move. It was noticed that after 15 – 20 minutes the hovercraft began to move at a very slow rate because of the battery powered being drained very quickly. It was also tested how the hovercraft would move with and without the battery compartment cover. It was deemed that the hovercraft moved faster with the battery compartment on as it holds the battery in place as it was noticed that when the hovercraft was switched on the batteries would be moving (as in coming out of the holder).

Hovercraft with the battery compartment cover on - https://drive.google.com/file/d/1l2z3n_Z4ZOBfJyxkESqjygHjZbbecrVF/view

Hovercraft with the battery compartment cover off –

https://drive.google.com/file/d/1-7UWdMDE_i0_StlAzXNzuWABtanbI-La/view

Conclusion and Recommendations

The aim of this project was to design and build (design using software such as AutoCAD and Fusion360) a small working model of a hovercraft with a system that allows the hovercraft to propel horizontally covering a distance of four yards. In addition, research and study were done to better understand the concept of the hovercraft which uses a horizontal and vertical propulsion system. The objectives/requirements of this project were that:

1. The hovercraft should be able to cover four yards or more.
2. Should be designed where it would be able to fit in a person's hand.
3. Hovercraft must be designed so that it could travel on land surface.
4. Must be designed so that be able to produce enough air cushion for it to over and produce enough thrust to move along the ground by itself.
5. Made of simple materials (Materials of plastic or foam etc.).
6. Budget of $50.

The expected outcome of this project was achieved, that is, design and build a small working model of a hovercraft with a system that allows the hovercraft to propel horizontally covering a distance of four yards. The hovercraft was tested and it works by producing lift and thrust forces which make the model move by itself without apply any other external forces.

Recommendations:

1. Ensure that the materials you are using are strong enough and durable

2. Ensure the most important parts of the hovercraft such as the skirt is properly made with the right material.

3. Ensure that you have strong knowledge of wire connection before attempting to wire anything.

4. Ensure the model that you build uses strong enough batteries for it to work.

Reference

wrike .com, “What are constraints in project management?” *Versatile & Robust Project Management Software*, 01-Jan-2006. [Online]. Available: https://www.wrike.com/project-management-guide/faq/what-are-constraints-in-project-management/. [Accessed: 20-Feb-2022].

S. Mathew, “Hovercraft,” *Mepits*, 24-Jun-2015. [Online]. Available: https://www.mepits.com/project/306/diy-projects/hovercraft. [Accessed: 23-Mar-2022].

torch templates, “Action plan templates free,” *Action Plan Templates Free | Torch Templates*, 01-Mar-2012. [Online]. Available: https://www.torchtemplates.net/2021/03/action-plan-templates-free.html. [Accessed: 20-Feb-2022].

W. .com, “Hovercraft,” *Wikipedia*, 13-Feb-2022. [Online]. Available: https://en.wikipedia.org/wiki/Hovercraft. [Accessed: 20-Feb-2022].

T. W. I. Global, “What is simulation? what does it mean? (definition and examples),” *TWI*, 08-Jun-2016. [Online]. Available: https://www.twi-global.com/technical-knowledge/faqs/faq-what-is-simulation. [Accessed: 20-Feb-2022].

Wright, P. H. (2002). *Introduction to engineering*. Wiley.

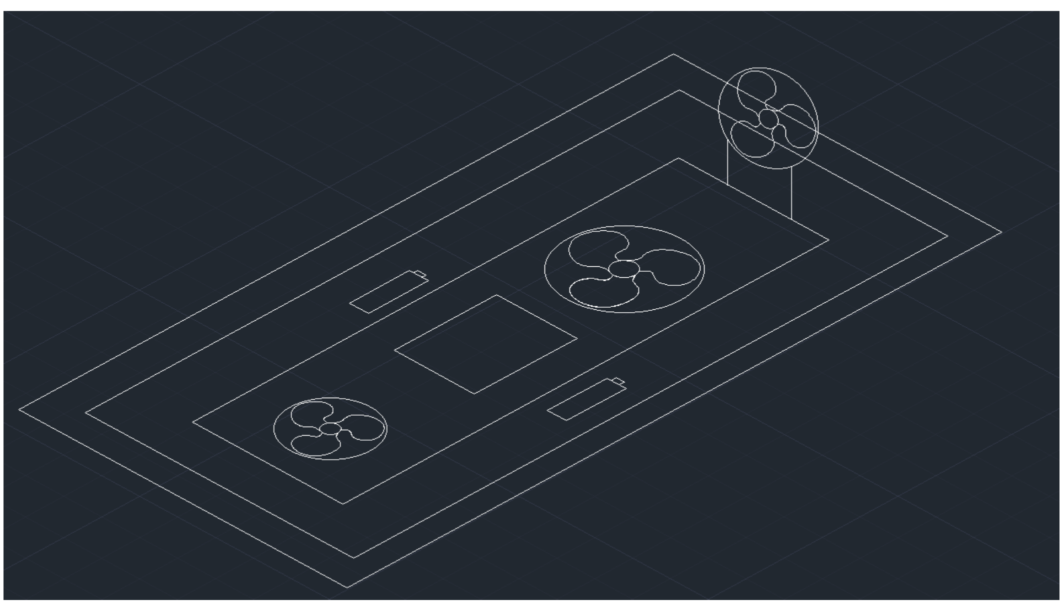
Öchsner Andreas. (2018). *Engineering applications for new materials and technologies*. Springer International Publishing.

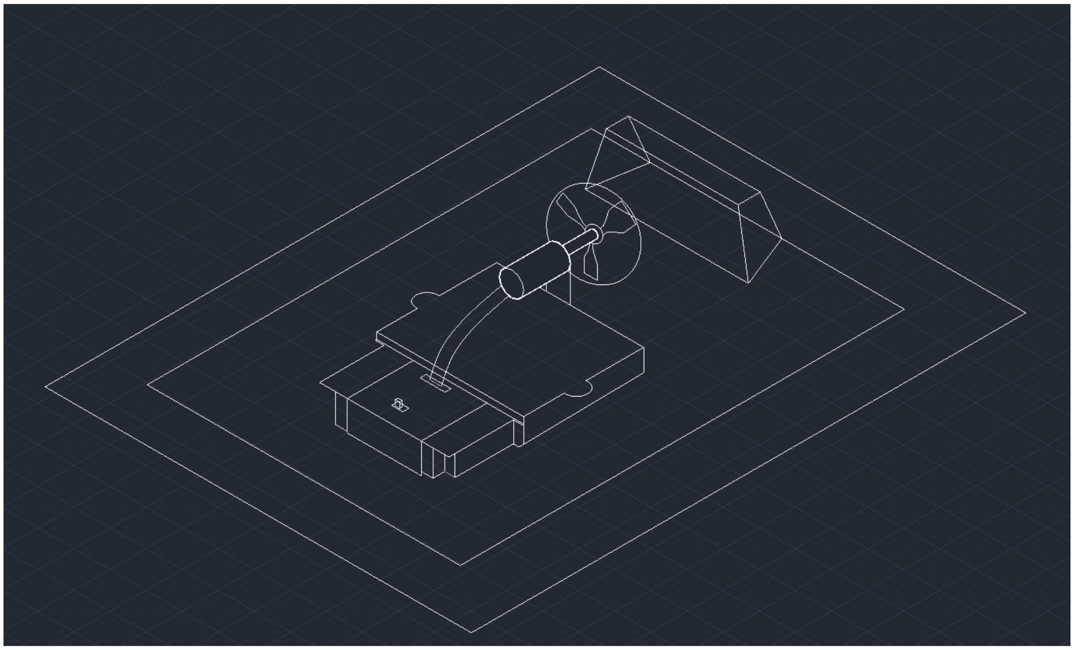
Appendixes

Table of Data used in project

Evaluation Matrix:	Possible Effectiveness Score	Score for Design 1	Score for Design 2
1. Cost Effectiveness	40	25	30
2. Weight	20	10	10
3. Travels 4 – 6 ft	20	10	15
4. Simplicity	10	10	5
5. Power Consumed	10	8	10
Total effectiveness score	100	63	70

Designs:





Materials and Components:

1. Base panel (Black Cardboard)

2.Plastic Skirt

3. Duct Template (Plastic)

4. Battery Compartment with switch and wires

5. Battery Compartment Cover

6. Motor Stand

7. Motor Cover

8.Attachment Bar

9. Propeller (Fan Blade)

10. Motor with wires

11. Screws.

12. Terminal Cap.

13. Double-sided Adhesive Tape.